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K2-Think: A Parameter-Efficient Reasoning System

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We introduce K2-Think, a reasoning system that achieves frontier performance with just a 32B parameter model — surpassing or matching much larger models such as GPT-OSS 120B and DeepSeek v3.1. Built on the Qwen2.5 base model, our system demonstrates that smaller models can compete at the highest levels through synergistic combination of advanced post-training and test-time computation techniques. Our approach is built on top of six key technical pillars: Long Chain-of-thought Supervised Finetuning, Reinforcement Learning with Verifiable Rewards (RLVR), Agentic planning prior to reasoning, Test-time Scaling, Speculative Decoding, and Inference-optimized Hardware, using only publicly available open-source datasets. K2-Think prioritizes mathematical reasoning, achieving state-of-the-art scores on public benchmarks for open source models, while also maintaining strong performance on other domains such as Code and Science. Our results validate that a more parameter-efficient model like K2-Think 32B can rival state-of-the-art systems through an integrative post-train recipe including long chain-of-thought training and strategic inference-time enhancements, paving the way for more accessible and affordable open-source reasoning systems. We have made K2-Think freely available at k2think.ai demonstrating best-in-class inference speeds, through the Cerebras Wafer-Scale Engine, delivering upwards of 2,000 tokens per second per request.

K2 Think

🤗 K2-Think (Model)	huggingface.co/LLM360/K2-Think
K2-Think (Code)	github.com/MBZUAI-IFM/K2-Think-SFT github.com/MBZUAI-IFM/K2-Think-Inference
K2-Think (Web)	k2think.ai

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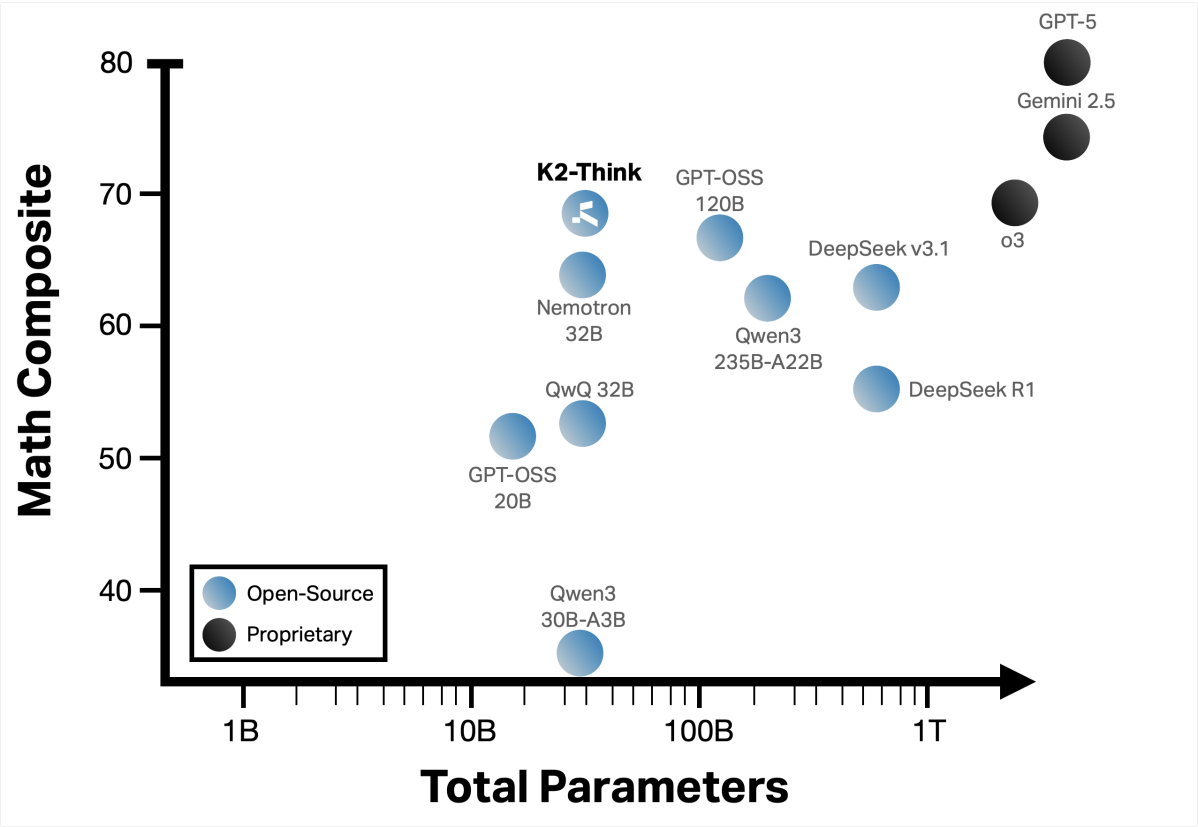

Figure 1: K2-Think exhibits remarkable parameter efficiency, providing comparable or superior performance to frontier reasoning models in complex math domains with an order of magnitude smaller model. The composite score here is the micro-average for each model over four complex math benchmarks, weighted by the number of questions in each benchmark (AIME 2024, AIME 2025, HMMT 2025, and Omni-MATH-HARD; see Section 3 for the benchmark details). Note: parameter counts for proprietary models are speculative.

1 Introduction

Recent advances in frontier reasoning models have highlighted the effectiveness of long chain-of-thought reasoning, enabled by large-scale supervised fine-tuning and reinforcement learning. Systems like OpenAI-O3 (OpenAI, 2025) and Gemini 2.5 (Google DeepMind, 2025) have achieved strong results on competition-level math benchmarks, complex coding tasks, and advanced scientific reasoning datasets, setting new milestones for reasoning-centered language models. These developments have also stimulated further exploration in the open-source community, where researchers have trained competitive reasoning systems with reinforcement learning (Yu et al., 2025; Hu et al., 2025; Wang et al., 2025c) and investigated mechanisms by which RL improves reasoning (Zeng et al., 2025; Yue et al., 2025; Shao et al., 2025; Agarwal et al., 2025b; Wang et al., 2025b).

In this report we introduce K2-Think: a competitive reasoning system built from the open-weight Qwen2.5-32B base model (Yang et al., 2024a). We break down our system into stages, including post-training and test-time, where an integrative recipe involving 6 major technical innovations was introduced over all stages spanning finetuning, reinforcement learning, planning, and hardware optimization to boost the base model reasoning capability, and we evaluated how each stage affects performance. **These components combine to enable a model of merely 32 billion parameters, with modest test-time compute, to match the mathematical reasoning performance of proprietary frontier models**. In fact, K2-Think emerges as the top open-source model for complex math benchmarks matching or exceeding previously leading

models that are orders of magnitude greater in size. Figure 1 presents a plot of the global micro-average of performance (essentially dividing the total number of correct answers by the total number of questions across all test sets) of each model over four challenging math competition tasks with respect to the total number of parameters for each model. The prominent positioning of K2-Think in the top-left visually depicts its superior parameter efficiency, demonstrating that it achieves State-of-the-Art performance among open-source models with a significantly smaller total parameter count. Detailed results and discussion for the benchmarks are presented in Section 3.

More specifically, K2-Think incorporates six key innovations to deliver a strong reasoning system. We first extend the base model with *chain-of-thought capabilities through Supervised Fine-tuning (SFT)*, followed by *Reinforcement Learning with Verifiable Rewards (RLVR)* to strengthen reasoning performance. We then enhance the model with inference-time techniques: *agentic planning* and *test-time scaling* using *Best-of-N sampling*. Finally, we deploy K2-Think with two speed optimizations: *speculative decoding* and Cerebras' Wafer-Scale Engine, an *inference-optimized hardware* system. This final stage enables the model to deliver its powerful chain-of-thought reasoning capabilities with near-instantaneous response times, deployed at speeds upwards of 2000 tokens per second per user request.

With the release of K2-Think, we share our experience and make available an important advancement in open-source language modeling, that aggressive post-training engineering and test-time computation, even with a modest commodity pretrained base model, can significantly boost reasoning capabilities in a cost-effective manner. Prior studies have reported that, in certain regimes, allocating more computation during inference can be more cost-effective than scaling model size (Snell et al., 2025); for recent frontier systems—including OpenAI's o1/o3 (OpenAI, 2024, 2025), DeepSeek-R1 (Guo et al., 2025), Google's Gemini 2.5 (Google DeepMind, 2025), and xAI's Grok4 (xAI, 2025) — model capabilities have been claimed to improve with increased test-time budgets (Ji et al., 2025a; Yang et al., 2025b).

In addition to releasing code and model weights, **we offer K2-Think through a public website and as a production-ready API endpoint.**[1] This allows the community to engage directly with a living system, shifting the emphasis from static artifacts to a deployable, studyable service that can be stress-tested and iterated on in the open. As dynamic inference-time reasoning becomes more complex, our API demonstrates the requirements of sophisticated systems for top performance, and provides an operational deployment delivering robustness, safety, and efficiency under real-world constraints.

In Section 2 we describe the development process and deployment of K2-Think, using the Cerebras Wafer-Scale Engine. Section 3 presents a thorough set of evaluations and ablations that attribute gains across post-training and test-time computation. Section 4 situates our contributions within the literature. We conclude in Section 5 with a summary overview, discuss our motivations for deploying this model, and chart future directions for extending reasoning performance with openly released models and deployment-ready systems.

2 K2-Think Development

We initiate K2-Think's development to study a complete post-training recipe for enhanced reasoning and establish best practices for extending our in-house foundation models. Throughout this study, we seek to validate published best practices as well as test original test-time computation ideas.

We choose to fine-tune a 32B-scale base model for K2-Think, as:

(1) it allows for fast iteration while providing strong base capabilities and

(2) its size suits both research and consumer computation frameworks.

[1] available upon request

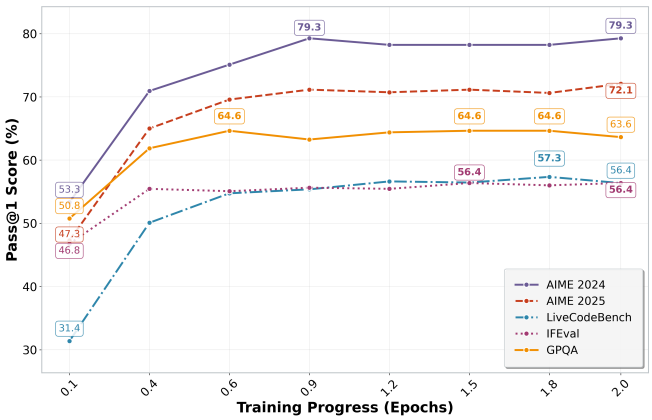

Figure 2: Pass@1 performance over training. Pass@1 of K2-Think-SFT across five benchmarks; the x-axis is training progress (epochs), the y-axis is pass@1 score.

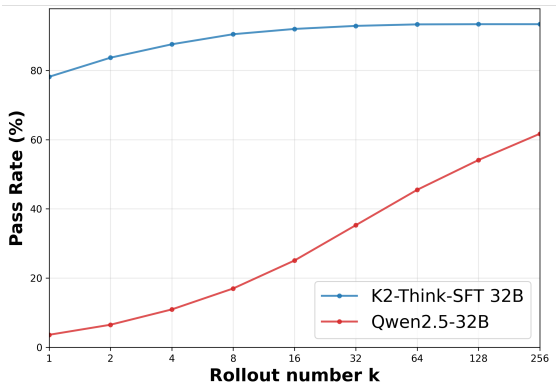

Figure 3: Pass@k on AIME2024. Pass@k of K2-Think-SFT and the Qwen-2.5 32B base model; the x-axis is the number of rollouts per question, the y-axis is pass rate.

Specifically, we selected Qwen2.5-32B as it is not tuned for reasoning, allowing us to fully validate our recipe's effectiveness.

2.1 Phase 1: Supervised Fine Tuning

The initial stage of K2-Think development constitutes supervised fine-tuning (SFT) of the base model using curated long chain-of-thoughts (CoT), establishing the first pillar of our complete reasoning system. This follows the paradigm introduced by DeepSeek in the development of their R1 model (Guo et al., 2025). This phase of training serves to provide guidance to the pre-trained base language model for generating structured responses to complex queries. Additionally, the model is trained to adopt an expected output format in which the model's reasoning process is made clear prior to producing an answer. By providing a token-by-token supervisory signal through extended CoT, the base model's intrinsic computation capabilities are expanded substantially (Wei et al., 2022; Schuurmans et al., 2024).

Our SFT phase[2] uses the existing `AM-Thinking-v1-Distilled` dataset,[3] composed of CoT reasoning traces and instruction/response pairs, with prompts drawn from tasks spanning mathematical reasoning, code generation, scientific reasoning, instruction following, and general chat (Ji et al., 2025b; Tian et al., 2025). In what follows, we will refer to this supervised fine-tuned model as **K2-Think-SFT**.

2.1.1 Observations

Our SFT experiments on Qwen2.5-32B yield several practical insights. Of particular note, we conduct a step-wise evaluation of K2-Think-SFT across five representative benchmarks. As shown in Figure 2, performance improves rapidly within the first third of training (roughly 0.5 epoch), particularly on mathematics benchmarks (AIME 2024 and AIME 2025). After this sharp initial gain, most benchmarks plateau, with AIME 2024 stabilizing around 79.3% pass@1 and AIME 2025 around 72.1%. GPQA and IFEval continue to exhibit modest upward trends, while LiveCodeBench shows a slower but steady improvement up to 56.4%. We observe that our SFT phase has reached convergence, with the model exhibiting diminishing returns to continued training on the dataset.

Apart from pass@1 scores, we also use pass@k to quantify reasoning performance under a fixed sampling budget k. Interpreting the pass@k curve as a capability boundary, we evaluate K2-Think-SFT. In Figure 3, assessing performance on AIME2024, our SFT model dominates the base model across sampling budgets. The SFT curves saturate near 93.3% by $k \approx 128$, whereas the base model continues to improve but remains

[2] Code, based on LLaMA-Factory, for SFT can be found at https://github.com/MBZUAI-IFM/K2-Think-SFT
[3] https://huggingface.co/datasets/a-m-team/AM-Thinking-v1-Distilled